Interim Submission: AI-Assisted Triage and Decision Support

<u>Paper 1:</u>

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus, 16*(5), e59906. https://doi.org/10.7759/cureus.59906

Summary:
This study examined how artificial intelligence is used in emergency department triage to improve patient prioritization and resource allocation. The authors conducted a scoping review of 29 studies published between 2018 and 2023 focusing on AI and machine learning applications in triage. Findings showed that AI models often outperform traditional triage systems in predicting hospitalization, critical illness, and ICU admission. However, limitations include concerns about algorithm transparency, data quality, ethical issues, and the need for broader validation across different healthcare settings.

<u>Paper 2:</u>

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics, 197*, 105838. https://doi.org/10.1016/j.ijmedinf.2025.105838

Summary:
The review explored the benefits, challenges, and future directions of AI-driven triage systems in emergency departments facing overcrowding and inconsistent patient prioritization. The authors synthesized existing literature on AI-assisted triage technologies and their impact on emergency care. Results suggest that AI can improve triage accuracy, support faster decision-making, and enhance resource management in high-demand environments. However, barriers such as data bias, system integration, explainability, regulatory compliance, and clinician trust continue to limit widespread adoption.

<u>Paper 3:</u>

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-assisted multimodal triage system for health centers. *Procedia Computer Science, 251*, 430–437. https://doi.org/10.1016/j.procs.2024.11.130

Summary:
This study focuses on overcrowding in emergency departments and the need for improved triage decision-making. The authors developed and evaluated several machine learning models, including Support Vector Machines, Random Forests, Gradient Boosting, XGBoost, and a stacking model, using the Korean Triage and Acuity Scale (KTAS). The stacking model achieved the best performance, with approximately 80% accuracy in predicting triage levels. A key limitation is that the model was tested on a specific dataset and triage framework, requiring further validation across different healthcare environments before general use.

Problem Statement:

Across the literature, emergency departments face increasing pressure from overcrowding, rising patient volumes, and limited healthcare resources, making accurate and timely triage difficult. Traditional triage methods rely heavily on clinician judgment and standardized protocols, which may lead to inconsistencies in patient prioritization. Studies consistently show that artificial intelligence can improve triage accuracy, predict patient outcomes, and support healthcare decision-making. However, concerns regarding transparency, bias, reliability, clinician trust, and integration into existing workflows remain unresolved. Therefore, further investigation is needed to evaluate the effectiveness and practical implementation of AI-assisted triage and healthcare decision-support systems in real-world clinical settings.